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& \max \quad c^\top x \\
& \text{s.t.} \quad Ax + s = b + 0 \\
& \quad \quad x, s \geq 0
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$$\begin{aligned}
& a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 \\
& \quad + a_{i4}x_4 + \cdots + a_{im}x_m \leq b_i \quad \forall i \in \llbracket n \rrbracket
\end{aligned} \tag{1}$$

$$\mathbf{x} \geq \mathbf{0} \tag{2}$$

$$\begin{aligned}
& x_i \in \{0, 1\} \quad \forall i \in \llbracket M \rrbracket \\
& \sum_{i=1}^m x_i \leq k
\end{aligned}$$